

Anas MADANI

Full Stack Software Engineer

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PROFILE

Software engineer graduated from ENSIAS (Software Engineering), full stack from backend to frontend, with a DevOps practice of deploying on Linux: continuous integration and delivery (CI/CD), containerization and orchestration. Experience in demanding environments (banking, datacenter), used to delivering tested, reviewed and documented code within an agile team.

EDUCATION

- **École Nationale Supérieure d'Informatique et d'Analyse des Systèmes (ENSIAS)** Rabat, Morocco
Engineering Degree – Software Engineering 2022 – 2025
- **CPGE – Settati** Settati, Morocco
Preparatory Classes for Engineering Schools, TSI 2019 – 2022

PROFESSIONAL EXPERIENCE

- **Banque Centrale Populaire (BCP)** Casablanca, Morocco
Software Engineer Intern (Graduation Project) February 2025 – July 2025
 - Independently designed and delivered a banking product recommendation platform: 8 Spring Boot microservices, a FastAPI recommendation API and a React/TypeScript frontend, deployed internally and presented to business teams.
 - Built the recommendation engine end-to-end: prepared 17,000 customer-product pairs, customer segmentation via K-Means, benchmarked 4 classification algorithms against a graph-based collaborative filtering approach (Neo4j GDS), with a Precision@K / Recall@K / NDCG@K evaluation framework.
 - Industrialized the platform: Docker containerization, event-driven communication with Kafka, Redis caching, Keycloak authentication with centralized JWT validation at the API Gateway level.
 - *Stack:* Java, Spring Boot/Cloud, React, TypeScript, Python, FastAPI, Neo4j, PostgreSQL, Redis, Kafka, Keycloak, Docker, Git
- **N+ONE DATACENTERS** Casablanca, Morocco
Software Engineer Intern August 2024 – September 2024
 - Built a tender/RFP management platform (Django REST Framework / React), covering the full consultation lifecycle, candidate submissions and a public portal for external companies.
 - Implemented security mechanisms: AES-256-GCM encryption for sensitive documents, role-based access control, JWT authentication with token rotation, operation audit logging and rate limiting on public endpoints.
 - Containerized the stack with Docker Compose (Django, PostgreSQL, React, Nginx reverse proxy), set up a GitHub Actions CI pipeline (flake8, pytest against PostgreSQL) and a scheduled deadline-monitoring service.
 - *Stack:* Python, Django REST Framework, React, Vite, PostgreSQL, Docker, Nginx, GitHub Actions, JWT, AES-GCM, pytest
- **Commune Er-rich** Er-rich, Morocco
Software Engineer Intern July 2023 – August 2023
 - Built a workforce management application for the municipality: staff profiles, absence tracking and automated payroll calculation, replacing procedures previously kept on paper. Sole developer on the project.
 - Full stack delivery: Spring Boot backend services, React and Tailwind CSS interfaces, PostgreSQL persistence.
 - Designed the relational database: data dictionary, functional dependencies and conceptual model, based on requirements gathered from the relevant department.
 - *Stack:* Java, Spring Boot, React, Tailwind CSS, PostgreSQL, UML, Git

PROJECTS

- **Practice Debt** (*Aug – Sep 2026*): Codeforces practice tracker that surfaces problems abandoned mid-contest and techniques gone stale, ranked in a single queue. Reverse-engineered the contest scoring and Elo-style rating formulas, quantified the model's error against all 20,489 rating changes in a single Codeforces round, and derived costs as model-vs-model differences so the residual bias cancels. Spring Boot 3 / PostgreSQL backend, React 19 (TypeScript) frontend, single-container deployment, 90+ tests. [GitHub](#)
- **BacSurv** (*Aug 2026*): Exam supervision planning for the Moroccan baccalaureate: assigns teachers to sessions under availability, workload and conflict constraints. Third build of a problem first attempted in 2025 – a TypeScript prototype, then a FastAPI rewrite, now Java/Spring Boot with a constraint solver. React 19 (TypeScript) frontend, Word/Excel export and a bilingual Arabic/French interface. [GitHub](#)
- **BusConnect** (*Oct 2024 – Jan 2025*): Bus reservation platform on a microservices architecture with Spring Boot 3, Spring Cloud (Eureka, API Gateway), React 18 and Kubernetes deployment. Integrates PostgreSQL, MySQL and MongoDB depending on the service. [GitHub](#)
- **MLOps Pipeline Generator** (*Oct 2024 – Jan 2025*): MLOps code generator based on Model-Driven Engineering (MDE) using Eclipse Ecore and Acceleo. Automatically generates Python scripts, Dockerfiles and Kubernetes configurations from metamodels via an SWT/EMF Forms interface. [GitHub](#)

SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, C, C++, SQL
Backend: Spring Boot, Spring Cloud, FastAPI, Django REST Framework
Frontend: React, TypeScript, Tailwind CSS, Vite
Databases: PostgreSQL, MySQL, MongoDB, Neo4j, Redis
DevOps & Infra: Docker, Kubernetes, Kafka, GitHub Actions, Nginx, Keycloak, Linux
Spoken Languages: Arabic (native), French (fluent), English (fluent)